

CODE SPORT



In this month's column, we continue our discussion on the problem of content moderation using NLP techniques, focusing on detecting hate speech in social media content.

In last month's column, we had discussed the problem of content moderation. Everything posted on social media sites such as Facebook or YouTube needs to be analysed in order to identify unsuitable or abusive content and ensure that it is taken down quickly. This includes filtering of inappropriate content such as pornography, gory visuals, exploitative images of minors, sexual solicitation, racism, etc, from text, video and speech posted on the Internet. Manual moderation has two major problems. First, it cannot scale as the rate at which new content gets posted to the social media increases exponentially. Second, the very task of moderating content causes stress and psychological issues for the human moderators who need to view the objectionable posts or videos as part of their daily work. Given these problems, there has been considerable focus on applying natural language processing and machine learning techniques to classify objectionable content automatically.

While there are various categories of objectionable content that need to be identified on social media, one important category is hate speech. This can be typically defined as content that targets individuals or groups, based on their characteristics, and demonstrates a clear intention to incite harm or to promote hatred. Typically, such hate speech is directed against specific communities such as blacks, Muslims, Jews or refugees and the content is intended to promote hatred against them. An example taken from Facebook's guidelines on hate speech content is the following: "Niggers! They are filthy idiots." This is directed against the specific community of blacks and promotes hatred towards them. Hate speech can also be directed against a single individual: "Hope that bitch falls over and breaks her leg." While the above examples explicitly contain abusive words, hate speech may not always use explicitly abusive words. Consider this example taken from Twitter: "Assimilate? No they

all need to go back to their own countries. #BanMuslims Sorry if someone disagrees, too bad."

The above example does not contain any abusive words, but still is a form of hate speech as it incites hatred against a specific community. Such forms of hate speech that do not contain abusive words are known as implicit hate speech. While explicit hate speech is easier to detect using lexicon based approaches, detecting implicit hate speech is extremely difficult, since it requires an understanding of the context.

A bigger challenge associated with hate speech detection is that it is extremely subjective. Consider this example: "Our class prom night just got ruined because u showed up. Who invited u anyway?" While some of the human moderators termed the above example as hate speech, a few others felt that it was simply sarcastic and not really inciting hatred towards an individual. Such subjectivity leads to considerable disagreement even among human moderators as to what constitutes hate speech, and this makes it even more difficult for machines to detect such speech.

It is also important to note that hate speech on social media consists of language that can contain slang or memes that are rapidly evolving and are highly contextual. Rapid evolution of social language and events makes it difficult to detect racial slurs. For example, consider this post: "He is the next Nikolas Cruz!". Here, the reference is to the recent Parkland School shooting in Florida, (https://en.wikipedia.org/wiki/Stoneman_Douglas_High_School_shooting) where Nikolas Cruz is the alleged shooter. This becomes a major challenge for automated techniques since they need to be updated frequently with evolving social media slang and with current affairs. In addition, many of the terms/slurs used in hate speech are culture and language specific. This requires that automated systems be localised or tuned to the cultural and social context. While humans have